Interpretable Sparse Dictionary Learning Approach for Multi-Class Abnormality Classification and Grounded Localization using 3D Chest CT

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Abstract

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Acronyms

AC	-	Alternating Current
DC	-	Direct Current
LV	-	Low Voltage
MV	-	Medium Voltage

Chapter 1

Introduction

Developing a unified architectural pipeline for multi-class abnormality classification and grounded localization in 3D Computed Tomography (CT)a ddresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is held accountable and its reasoning aligns with clinical diagnosis.

1.1 Background

The transition from 2D projectional radiography to volumetric CT represents a paradigm shift in medical image analysis. For Lung disease diagnosis, Chest X-rays (CXR) are commonly used as the first imaging test due to its low cost and low radiation exposure. However, studies show that their capability to detect abnormailities is limited when compared to 3D CT. [why?].

CXR has historically benefited from the availability of massive, radiologist-annotated datasets, while the analysis of 3D chest CT volumes has been severely constrained by a lack of such data. However, the recent introduction of large-scale, text-paired datasets has facilitated the development of state-of-the-art (SotA) vision-language foundation models tailored specifically for volumetric imaging. (citations for ReX-GroundingCT, RadGenome-Chest CT, RAD-ChestCT)[][][].

Beyond deep learning approaches, the Label Consistent K-SVD (LC-KSVD) algorithm offers a viable approach for interpretable CT analysis. LC-KSVD is a dictionary learning algorithm that incorporates label information directly into the dictionary construction process [lc-ksvd citation], making it inherently discriminative and well-suited for multi-abnormality tasks.

1.2 Problem Statement

The clinical utility of Deep Learning-based approaches is hindered by a lack of interpretability. Tools like Grad-CAM provide visual heatmaps highlighting important regions in an input image by analyzing gradients from the final convolutional layer[citation]. These post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this project is to address the limitations of existing Deep Learning based approaches for Lung abnormality classification and localization by developing an inheerntly interpretable dictionary learning framework based on LC-KSVD. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning approach to learn normal and pathological dictionary atoms using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.
- Benchmark model performance against state-of-the-art classification and segmentation models on the same dataset.
- Evaluate abnormality localization by the model on a real-world dataset with expert radiologist review.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

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1.4 Literature Review

The landscape of 3D CT analysis is currently divided into three primary architectural paradigms:

- Classification-based models that prioritize diagnostic labels
- Segmentation-based models that prioritize boundary precision
- Unified architectures designed for grounded localization

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SotA models are predominantly specialized in either classification or segmentation. However, recent systems, particularly in 2025 and 2026, are moving toward multi-task learning or foundation models that can perform both diagnosis and voxel-level segmentation simultaneously.

The efficacy of these models is increasingly evaluated on large-scale, multi-institutional benchmarks such as CT-RATE, FLARE 2024-2025 and MSWAL, which provide the complexity necessary to test generalizability across diverse patient populations and scanner manufacturers.

1.4.1 Classification Architectures

Classification-based models for 3D CT have shifted away from 3D Convolutional Neural Networks (CNNs) such as 3D U-Net, 3D ResNet and DenseNet pipelines, toward Vision Transformers (ViTs) and self-supervised encoders. These models are designed to identify the presence of one or more abnormalities within a full volumetric scan. The benchmark for this paradigm is the CT-RATE dataset, which introduced the first public large-scale 3D medical imaging dataset with paired textual reports, enabling the development of Contrastive Language-Image Pretraining for Computed Tomography (CT-CLIP).

(About CT-CLIP) Recent benchmarks such as Merlin and DINOv3 have further challenged the necessity of domain-specific pretraining.

Merlin is a vision-language foundation model exclusively pre-trained on abdominal CT scans. Despite it's training, Merlin has consistently demonstrated exceptional zero-shot performance on chest CT classification tasks, even when it was restricted to a linear probe, significantly outperforming models like CT-CLIP, M3FM, and CT-FM that were explicitly trained on thoracic data, by an average of 12.3% in Area Under the Receiver Operating Characteristic Curve (AUROC). Specifically, in cross-domain retrieval tasks, Merlin surpassed CT-CLIP by 24.7%. DINOv3, a self-supervised ViT pretrained on over 1.7 billion natural images, outperform specialized medical models like CT-Net and CT-CLIP across all metrics on several classification tasks due to it's ability to capture fine-grained textural features.

1.4.2 Segmentation Architectures

The prevailing SotA for supervised volumetric segmentation remains variations of the 3D U-Net, such as nnU-Net, which utilizes an adaptive framework to optimize preprocessing and architecture for specific datasets. However, the field has recently shifted toward foundation models capable of universal segmentation such as SegVol and SAM-Med3D. The SegVol model is the first 3D foundation model specifically designed for universal volumetric medical image segmentation. Similarly, SAM-Med3D adapts the Segment Anything Model (SAM) for 3D medical data.

Recent evaluations published at the Machine Learning for Healthcare (MLHC) Conference in 2025, which specifically utilized the ReXGroundingCT dataset, revealed that the current SotA text-prompted segmentation models that demonstrate near-perfect performance in general computer vision tasks struggle with the nuanced clinical descriptions and ill-defined boundaries of chest CT abnormalities.

1.4.3 Unified Architectures

Recent models have been designed to integrate abnormality detection, segmentation, and multimodal chain-of-thought reasoning, enabling pixel-level localization, structured report generation, and physician-like diagnostic inference in a single framework. BiomedParse, and it's production-ready, version MedImageParse developed by Microsoft, are foundation models designed to parse 3D medical images. Other notable models include Citrus-V, MTMed3D and MedVista3D, where each architecture incorporates a multi-task learning (MTL) framework.

1.4.4 Explainability Approaches

Grad-CAM
Seg-Grad-CAM
Hires-CAM

1.5 Report Outline

Chapter 2

Background

2.1 LC-KSVD

The LC-KSVD algorithm solves an optimization problem that balances reconstruction accuracy with label consistency and classification error. Given a set of image patches P, the goal is to learn a dictionary D and sparse coefficients X that minimize the following objective:

$$\min_{D,X,A,W} \|P - DX\|_F^2 + \alpha\|Q - AX\|_F^2 + \beta\|H - WX\|_F^2 \quad \text{s.t.} \quad \forall i, \|x_i\|_0 \leq T$$

The first term $\|P - DX\|_F^2$ is the standard K-SVD reconstruction error. The second term $\|Q - AX\|_F^2$ introduces the "discriminative sparse-code error," where Q is a label-consistent constraint matrix that encourages patches from the same class to activate the same dictionary atoms. The third term $\|H - WX\|_F^2$ is the classification error, where W is a linear classifier matrix.

This formulation ensures that the dictionary atoms are not just representative of the image data but are also class-specific. In 3D CT multi-abnormality segmentation, this allows the model to learn distinct "sub-dictionaries" for each abnormality, facilitating easier classification of individual voxels.

Chapter 3

Methodology

3.1 Implementation of the LC-KSVD Approach

3.1.1 Developing the Python library

3.1.2 Back-Projecting Class-Discriminative Atoms

The unique intuition we propose in our approach is to generate spatial contribution maps through back-projection. Because each atom in the dictionary D is associated with a specific class label through the matrix Q, the model can theoretically "explain" its decision for a test patch by examining which atoms were used in its sparse representation.

To generate a spatial contribution map for a class c, the algorithm identifies all atoms d_k that have been assigned to that class. For a query voxel or patch with sparse coefficients x, the contribution C_c is calculated as the partial reconstruction:

$$C_c = \sum_{k \in \text{Class } c} d_k x_k$$

This process should theoretically back-project the class-specific features into the original voxel space, creating a map that highlights the regions contributing to the detection of a specific abnormality. for class-discriminative atoms operating on raw voxel-space inputs, the back-projection is an exact, closed-form spatial attribution.

3.1.3 Deriving Explainability through Segmentation Masks

To evaluate the clinical accuracy of LC-KSVD, these contribution maps are converted into binary segmentation masks for comparison with ground truth annotations. This conversion can be achieved through various thresholding methods.

Grad-CAM's approximation has two sources of imprecision — the gradient itself is a local linear approximation of a non-linear function, and the spatial upsampling from the coarse feature map to input resolution introduces further interpolation error. LC-KSVD's contribution map has only one source of imprecision: the thresholding step you apply to binarize it into a mask. The map itself — before thresholding — is exact given the sparse code and the dictionary.

Unlike the passive Grad-CAM approach, atom back-projection is generative. The atoms are what the model actually used to reconstruct and classify the signal. There's no proxy involved. The localization evidence is produced by the same computation that produced the classification, not by a separate post-hoc analysis run afterwards.

3.2

Chapter 4

Experimentation and Evaluation

4.1 Project Proposal

4.2 Project Progress

4.3 End Exam

Chapter 5

Discussion

Appendix A

Appendix 1

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

Appendix B

Appendix 2

Bibliography